Dinosaurs: A Field Guide to Biology, Behavior and Extinction

Internal Training Edition, Revision 3.1. Compiled for retrieval experiments; values are approximate published estimates.

1 Introduction and Scope

This field guide summarizes current knowledge about non-avian dinosaurs for internal training. It covers the timeline of the Mesozoic Era, the major groups, profiles of ten well-known species, and the main topics of dinosaur biology: diet, reproduction, metabolism, behavior and senses. It closes with the end-Cretaceous mass extinction, the history of fossil discovery, and a list of open questions where earlier ideas have been revised.

Dinosaurs were a diverse group of reptiles that dominated land ecosystems for more than 160 million years, from roughly 230 million years ago (Ma) in the Late Triassic until 66 Ma at the end of the Cretaceous. Values given in this guide, such as body length and mass, are published estimates. Estimates differ between studies, sometimes by a factor of two, and should be read as approximate ranges rather than exact measurements.

2 Timeline of the Mesozoic Era

The Mesozoic Era is divided into three periods. Each period saw different dinosaur faunas, and many famous species that are often shown together never actually met.

2.1 Triassic Period (252-201 Ma)

The Triassic began after the end-Permian extinction, the largest mass extinction in Earth's history. The earliest known dinosaurs appear in rocks of about 230 Ma in Argentina and Brazil, for example Eoraptor and Herrerasaurus. These early forms were small to medium-sized bipeds and were not yet dominant; crocodile-line archosaurs were more common. A further extinction event at the end of the Triassic removed many competitors and opened ecological space that dinosaurs filled during the Jurassic.

2.2 Jurassic Period (201-145 Ma)

During the Jurassic, dinosaurs became the dominant large land animals. Giant long-necked sauropods such as Brachiosaurus and Diplodocus, plated stegosaurs and large predators such as Allosaurus lived in the Late Jurassic. The oldest widely accepted bird, Archaeopteryx, is known from Late Jurassic limestone in Germany dated to about 150 Ma. Stegosaurus lived about 155-150 Ma, so it is separated from Tyrannosaurus rex by more time than separates Tyrannosaurus rex from humans.

2.3 Cretaceous Period (145-66 Ma)

The Cretaceous was the longest period of the Mesozoic. Flowering plants spread, and new herbivore groups such as hadrosaurs (duck-billed dinosaurs) and ceratopsians (horned dinosaurs) diversified. The largest known land animals, titanosaur sauropods such as Argentinosaurus, lived in the mid-Cretaceous of South America. The period, and the age

of non-avian dinosaurs, ended 66 Ma with the K-Pg extinction described in Section 10.

3 Classification

Dinosaurs are traditionally divided into two major groups based on the structure of the hip (pelvis). The names are older than modern evolutionary studies and can be misleading, as explained below.

3.1 Saurischia

Saurischia means lizard-hipped. It contains two branches: theropods and sauropodomorphs. Theropods were mostly bipedal and mostly carnivorous, and include Tyrannosaurus rex, Velociraptor, Spinosaurus and Troodon. Sauropodomorphs include the long-necked sauropods such as Brachiosaurus and Argentinosaurus, which were quadrupedal herbivores and include the largest land animals ever known.

3.2 Ornithischia

Ornithischia means bird-hipped and contains only herbivores, many with beaks and specialized teeth. The group includes armored dinosaurs (Stegosaurus, Ankylosaurus), horned dinosaurs (Triceratops) and duck-billed dinosaurs (Parasaurolophus, Maiasaura). Despite the name, birds did not evolve from ornithischians.

3.3 Birds Are Living Dinosaurs

Birds evolved from small feathered theropods within Saurischia, so birds are classified as avian dinosaurs. Paleontologists therefore speak of non-avian dinosaurs when referring to the extinct groups. Discoveries in China since 1996, beginning with Sinosauropteryx, have shown that feathers or feather-like filaments were widespread among theropods and that many bird features, such as wishbones and air-filled bones, appeared first in non-avian dinosaurs.

4 Species Profiles

Table 1 lists approximate size, age and diet for the ten species profiled in this section. Lengths are in meters and masses in metric tonnes; Ma means million years ago.

Species	Period	Age (Ma)	Length (m)	Mass (t)	Diet	Region
Tyrannosaurus rex	Late Cretaceous	68-66	12	8	Carnivore	North America
Velociraptor	Late Cretaceous	75-71	2	0.02	Carnivore	Mongolia
Triceratops	Late Cretaceous	68-66	9	6-12	Herbivore	North America
Brachiosaurus	Late Jurassic	154-150	22	30-50	Herbivore	North America
Stegosaurus	Late Jurassic	155-150	9	5	Herbivore	North America
Spinosaurus	Mid Cretaceous	99-93	14-15	7	Piscivore	North Africa
Ankylosaurus	Late Cretaceous	68-66	6-8	5-8	Herbivore	North America
Parasaurolophus	Late Cretaceous	76-73	9.5	2.5	Herbivore	North America
Argentinosaurus	Mid Cretaceous	95-93	30-35	65-75	Herbivore	Argentina
Troodon	Late Cretaceous	77	2.4	0.05	Omnivore	North America

Table 1. Approximate size, age and diet of profiled species.

4.1 Tyrannosaurus rex

Tyrannosaurus rex lived in western North America at the very end of the Cretaceous, about 68-66 Ma, and was among the last non-avian dinosaurs. Adults reached about 12 m in length and 8 t in mass. Its bite force has been estimated at roughly 35,000 newtons, one of the highest of any land animal, and its teeth were thick and serrated for crushing bone. Bone histology shows that T. rex reached adult size around age 20 and that the oldest known individuals were about 28 years old.

4.2 Velociraptor

Velociraptor was a small dromaeosaurid theropod from Mongolia, about 75-71 Ma. It was roughly 2 m long, including a long stiff tail, and weighed about 15-20 kg, closer to a large turkey than to the human-sized animals shown in films. Quill knobs on a forearm bone, described in 2007, show that Velociraptor had feathers. A famous fossil preserves a Velociraptor locked in combat with a Protoceratops.

4.3 Triceratops

Triceratops was a large ceratopsian from North America, 68-66 Ma, living alongside T. rex. It reached about 9 m and 6-12 t. It had two long brow horns, a short nose horn and a solid bony frill. Healed bite marks from T. rex on Triceratops bones show that the two species interacted, and some injuries are consistent with horn-to-horn fighting between Triceratops individuals.

4.4 Brachiosaurus

Brachiosaurus was a sauropod from the Late Jurassic of North America, about 154-150 Ma. Its front legs were longer than its hind legs, giving it a sloping back and a raised neck, which suggests it fed high in trees. It was about 22 m long; mass estimates range from about 30 to 50 t depending on the method used.

4.5 Stegosaurus

Stegosaurus lived in the Late Jurassic of North America, about 155-150 Ma. It was about 9 m long and weighed around 5 t. Two rows of large bony plates ran along its back, and its tail ended in four spikes, informally called the thagomizer, used for defense. Its brain was notably small relative to its body size.

4.6 Spinosaurus

Spinosaurus lived in North Africa about 99-93 Ma and is the longest known theropod, with length estimates of about 14-15 m and mass estimates that vary widely, roughly 7 t. It had a tall sail on its back and a long snout with conical teeth suited for catching fish. How much time Spinosaurus spent in water is debated; see Section 12.

4.7 Ankylosaurus

Ankylosaurus was a heavily armored herbivore from North America, 68-66 Ma, about 6-8 m long and 5-8 t. Bony plates called osteoderms covered its back, and its tail ended in a heavy club of fused bone that could deliver strong blows to predators.

4.8 Parasaurolophus

Parasaurolophus was a hadrosaur from North America, about 76-73 Ma, about 9.5 m long and 2.5 t. A long hollow crest extended backward from its head. The crest contained air

passages connected to the nose, and computer models suggest it produced low-frequency sounds; see Section 8.

4.9 Argentinosaurus

Argentinosaurus was a titanosaur sauropod from Argentina, about 95-93 Ma. It is known from only a few bones, so its size is estimated by comparison with relatives: about 30-35 m long and 65-75 t, making it one of the largest land animals ever known.

4.10 Troodon

Troodon was a small, lightly built theropod from North America, about 77 Ma, about 2.4 m long and 50 kg. It had large forward-facing eyes and one of the largest brain-to-body ratios among non-avian dinosaurs. Its teeth suggest an omnivorous or flexible diet including small animals, eggs and possibly plants. Troodon fossils have been found from Alaska to the southern United States, indicating that it tolerated a wide range of climates.

5 Diet and Feeding

Dinosaur diets are reconstructed from tooth shape, jaw mechanics, stomach contents, fossil dung (coprolites) and tooth wear.

5.1 Carnivores

Most theropods were carnivores. Large tyrannosaurs had thick teeth for crushing bone, while smaller dromaeosaurs such as Velociraptor had blade-like teeth and an enlarged sickle claw on each foot for gripping prey. Coprolites attributed to T. rex contain crushed bone fragments, which shows that it ate bones as well as flesh. Large carnivores were probably both active predators and opportunistic scavengers.

5.2 Herbivores

Herbivorous dinosaurs used several strategies. Sauropods stripped leaves with peg-like teeth and swallowed them without chewing; large individuals may have eaten several hundred kilograms of plant material per day. Hadrosaurs and ceratopsians, in contrast, chewed their food using dental batteries: tightly packed columns of hundreds of teeth that were continuously replaced. Many herbivores were specialists that depended on particular plant groups.

5.3 Omnivores and Generalists

Some dinosaurs had flexible diets. Oviraptorosaurs, troodontids such as Troodon, and many early birds probably ate a mix of plants, seeds, insects, small vertebrates and eggs. Such generalist feeding allows an animal to switch food sources when one becomes scarce, a point that becomes important in Section 10.3.

6 Reproduction and Growth

All known dinosaurs laid eggs. Nesting sites in Montana revealed Maiasaura, the good mother lizard, with nests spaced about one adult body length apart and juveniles that stayed in the nest after hatching, suggesting parental care. Oviraptor fossils from Mongolia have been found sitting on clutches of eggs in a brooding posture similar to birds. Studies of embryonic teeth published in 2017 estimated incubation times of about

three months for Protoceratops and about six months for the hadrosaur Hypacrosaurus, much longer than for birds of similar size. A 2020 study suggested that the earliest dinosaurs laid soft-shelled eggs and that hard shells evolved several times. Growth rates varied widely: small theropods matured within a few years, while large sauropods grew for decades.

7 Thermoregulation and Metabolism

Whether dinosaurs were cold-blooded (ectothermic, relying on outside heat) or warm-blooded (endothermic, producing heat internally) was debated for decades. Growth-rate studies published in 2014 suggested that many dinosaurs were mesothermic, with metabolic rates between those of modern reptiles and mammals. Very large sauropods would have retained body heat simply because of their size, an effect called gigantothermy. Feathered theropods close to the bird line were probably fully endothermic and used feathers for insulation. Chemical analysis of eggshells published in 2022 suggested elevated metabolic rates in several groups, but the question is not settled for all dinosaurs.

8 Social Behavior and Communication

Parallel trackways show that some sauropods and hadrosaurs moved in groups. Bonebeds containing many individuals of the same species, for example of Allosaurus or of the tyrannosaur Albertosaurus, have been interpreted as evidence of group living, though some may result from animals gathering at drying water sources. Horns, frills, plates and crests probably served in display, species recognition and mate selection as well as in defense. The hollow crest of Parasaurolophus likely worked as a resonating chamber for low-frequency calls that could travel long distances.

9 Senses and Intelligence

Brain shape is studied using CT scans of the braincase. Tyrannosaurus had very large olfactory bulbs, indicating an excellent sense of smell, and forward-facing eyes that gave it overlapping fields of vision. Troodon had the highest encephalization quotient (brain size relative to expected brain size for its body mass) among known non-avian dinosaurs, comparable to some modern birds. Large sauropods and stegosaurs had small brains relative to body size. Intelligence in extinct animals cannot be measured directly, so these comparisons are rough.

10 Mass Extinction

The Cretaceous-Paleogene (K-Pg) mass extinction 66 Ma eliminated all non-avian dinosaurs, the flying pterosaurs, the large marine reptiles and about three quarters of all plant and animal species.

10.1 The Chicxulub Impact

In 1980, Luis and Walter Alvarez and colleagues reported a worldwide layer of clay enriched in iridium, an element rare in Earth's crust but common in asteroids, exactly at the K-Pg boundary. The matching impact crater, about 180 km wide, was identified in

1991 at Chicxulub on the Yucatan Peninsula of Mexico. The impact of an asteroid roughly 10 km across released debris and soot that blocked sunlight for months to years. Photosynthesis collapsed, global temperatures dropped sharply, an effect described as an impact winter, and food chains on land and in the oceans broke down.

10.2 The Deccan Traps

At about the same time, massive volcanic eruptions in India, the Deccan Traps, released large volumes of carbon dioxide and sulfur gases over hundreds of thousands of years. Some researchers argue that this volcanism stressed ecosystems before the impact; most current work considers the asteroid impact the main cause of the extinction, with volcanism as a possible contributing factor.

10.3 Who Survived and Why

The only dinosaurs that survived were birds, and not all birds survived. Survivors across many groups shared some traits: small body size, which allows faster reproduction and smaller food requirements; generalist diets, especially the ability to eat seeds, insects or detritus, which remained available when plants stopped growing; and the ability to shelter in burrows or water. Studies of bird beaks across the boundary suggest that toothless, seed-eating ground birds were more likely to survive than tree-dwelling species. Large specialized herbivores and the predators that depended on them were among the most vulnerable.

11 Fossil Discovery and Methods

The first dinosaur to be scientifically named was Megalosaurus, described by William Buckland in 1824, followed by Iguanodon in 1825. Richard Owen coined the name Dinosauria in 1842. Dinosaur fossils are now known from every continent, including Antarctica. Important sites include the Hell Creek Formation in Montana and the Dakotas, which preserves the last non-avian dinosaurs, the Gobi Desert of Mongolia, and the Liaoning Province of China, where fine-grained volcanic ash preserved feathers. Modern methods include CT scanning of skulls and eggs, bone histology to count growth rings, isotope analysis of teeth to study diet and climate, and analysis of pigment structures called melanosomes, which has allowed the colors of a few small feathered dinosaurs to be reconstructed.

12 Open Questions and Revised Hypotheses

Several popular ideas about dinosaurs have been revised as new evidence appeared. When older and newer claims conflict, this guide follows the more recent work, but readers should note that debates continue.

12.1 How Fast Could T. rex Run?

Older popular estimates gave T. rex running speeds of 60-70 km/h. Biomechanical studies since 2002 and a 2017 computer model concluded that its skeleton could not withstand such speeds and estimated a maximum of about 20-30 km/h for adults, fast walking rather than true running. Juveniles, being lighter, were probably faster.

12.2 Was Spinosaurus Aquatic?

Studies in 2014 and 2020 argued that Spinosaurus was a semi-aquatic animal that swam using a paddle-like tail. A 2022 study of its body proportions and bone density reached a different conclusion: that it waded in shallow water like a heron and hunted fish from the shore, rather than swimming. The question remains open.

12.3 Did Tyrannosaurs Have Feathers?

Early relatives of T. rex had feathers: Yutyrannus, described in 2012 from China, was about 9 m long and covered in long filaments. Skin impressions from T. rex and other large tyrannosaurs, however, show small scales. A 2017 study concluded that large tyrannosaurs were mostly scaly, though small patches of feathers cannot be ruled out.

13 Glossary

K-Pg boundary: the geological boundary between the Cretaceous and Paleogene periods, 66 Ma, marking the mass extinction; formerly called the K-T boundary.

Endothermy: producing body heat internally (warm-blooded). Ectothermy: relying on external heat (cold-blooded). Mesothermy: intermediate metabolism.

Gigantothermy: maintaining a stable body temperature because of very large body size.

Encephalization quotient (EQ): brain size relative to the brain size expected for an animal of the same body mass.

Osteoderm: a bony plate embedded in the skin. Coprolite: fossilized dung. Bonebed: a deposit with many fossil individuals.

Dental battery: a set of tightly packed, continuously replaced teeth forming a grinding surface.